Reporting No:02 Week No:02 From :28/07/26 To: 03/08/26
Project ID: 34
Project Title: SignBridge-AI

WEEKLY REPORT 2

Work done in last week (Attach supporting Documents):

Dataset Engineering, Preprocessing & Feature Preparation

1. Sign Language Dataset Research and Selection

- Researched publicly available sign-language datasets suitable for developing the SignBridge AI system.
- Studied different types of datasets, including:
  - Isolated sign/word-level datasets
  - Continuous sign-language datasets
  - Indian Sign Language datasets
  - Video-based datasets
  - Landmark-based datasets
- Compared datasets based on:
  - Number of classes
  - Number of videos
  - Number of signers
  - Dataset format
  - Language
  - Availability of annotations
  - Suitability for real-time recognition
- Identified suitable datasets for subsequent model development, including INCLUDE, WLASL and How2Sign.
- Studied the difference between isolated sign recognition and continuous sign-language recognition.

2. Dataset Structure and Video Analysis

- Downloaded/organized the selected dataset resources.
- Inspected the dataset directory structure and metadata.
- Studied how individual sign classes and video samples are represented.
- Examined video properties such as:
  - Resolution
  - Number of frames
  - Frame rate
  - Duration
  - Number of samples
- Created an organized local directory structure for raw and processed data.

  Example:

  data/

  ├── raw/

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```
├── processed/
├── landmarks/
├── train/
├── validation/
└── test/
```

3. Video-to-Frame Processing

- Developed preprocessing functionality for reading sign-language videos using OpenCV.
- Extracted individual frames from video sequences.
- Studied temporal information present between consecutive frames.
- Implemented basic preprocessing operations including:
  - Frame resizing
  - Frame extraction
  - Image format conversion
  - Frame selection
- Investigated the importance of maintaining temporal ordering of frames for future LSTM/Transformer-based sequence recognition.

The preprocessing pipeline was designed as:

Sign Language Video
↓
Frame Extraction
↓
Frame Preprocessing
↓
MediaPipe
↓
Landmark Extraction
↓
Numerical Sequence

4. MediaPipe Landmark Generation

- Extended the MediaPipe pipeline developed during Week 1 for processing sign-language videos.
- Applied landmark detection to individual video frames.
- Extracted relevant:
  - Left-hand landmarks
  - Right-hand landmarks
  - Pose landmarks
  - Facial landmarks where applicable
- Converted detected landmarks into numerical feature representations.
- Investigated missing landmark cases and detection failures.
- Stored extracted landmark sequences for future deep-learning model training.

The basic transformation achieved was:

Video
↓
Frames
↓

MediaPipe Holistic
↓
Hands + Face + Pose
↓
Landmark Coordinates
↓
Numerical Feature Sequence

5. Data Preprocessing and Training Dataset Preparation

- Studied methods for preparing landmark data for machine-learning models.
- Implemented/experimented with:
  - Coordinate normalization
  - Feature scaling
  - Sequence length handling
  - Missing-value handling
  - Label encoding
- Organized samples into training, validation, and testing categories.
- Ensured that the data maintains the correct temporal sequence required for sign-language recognition.
- Prepared the initial dataset pipeline for the upcoming LSTM/Transformer model-training stage.

Reason for incomplete work:

1.
Training of the final deep-learning sign-recognition model was not completed because Week 2 primarily focused on understanding, preprocessing, and preparing the dataset.
2.
Continuous sign-language recognition requires substantially more sequential data and preprocessing than isolated sign recognition. Therefore, additional dataset preparation and experimentation will continue alongside model development.
3.
Advanced features such as sentence formation, grammar correction, speech processing, multilingual translation, AI tutor, and meeting mode are planned for later development stages after the core sign-recognition model is established..

Plans for next week:

1. Model Architecture

- Study sequence-based deep-learning architectures.
- Implement a baseline LSTM model for sign recognition.
- Understand:
  - Input sequences
  - Hidden states

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- Time steps
- Output classes
- Loss functions
- Optimizers

2. Model Training

- Train the initial sign-recognition model.
- Experiment with:
  - Batch size
  - Learning rate
  - Number of LSTM layers
  - Hidden dimensions
  - Number of epochs
- Monitor:
  - Training accuracy
  - Validation accuracy
  - Training loss
  - Validation loss

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3. Model Evaluation

- Evaluate the trained model on unseen samples.
- Generate:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion matrix
- Identify frequently confused signs.
- Perform initial model improvements based on the evaluation results.

References:

1. MediaPipe Documentation
2. OpenCV Documentation
3. PyTorch Documentation
4. NumPy Documentation
5. Pandas Documentation
6. WLASL Dataset
7. INCLUDE – Indian Sign Language Dataset
8. How2Sign Dataset
9. Research literature on sign-language recognition and sequence modeling
10. Git and GitHub Documentation

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Signature of External Guide **Signature of Internal Guide**

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